

Second Problem Sheet

on

Linear Algebra

TA: Ronny Morales Carmona
Topic: Algebras

“Algebra is the offer made by the devil to the mathematician. The devil says: I will give you this powerful machine, it will answer any question you like. All you need to do is give me your soul.”
-Michael Atiyah

Problem 1

Show that the cross product $\times : \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by

$$\mathbf{x} \times \mathbf{y} = \epsilon_{ijk} x_i y_j \mathbf{e}_k, \quad \mathbf{x} = x_i \mathbf{e}_i, \quad \mathbf{y} = y_j \mathbf{e}_j,$$

with $\{\mathbf{e}_k\}_{k=1}^3$ is the canonical basis for $\mathbb{R}^3$, turns it into a nonassociative, noncommutative algebra.

Problem 2

The **centre** of an algebra $\mathcal{A}$, denoted by $\mathcal{Z}(\mathcal{A})$ is defined as the set of all elements of $\mathcal{A}$ that commute with all elements of $\mathcal{A}$. Show that the centre of an algebra is a subspace of that algebra. Moreover, show that if the algebra is associative, then its centre is a subalgebra.

Problem 3

An algebra is said to be **central** if its centre is $\mathcal{Z}(\mathcal{A}) = \text{span}\{\mathbf{1}\}$. Show that the algebra of quaternions is central.

Problem 4

Given an algebra $\mathcal{A}$ and a basis $B = \{\mathbf{e}_i\}_{i=1}^N$ for the underlying linear space, one can write

$$\mathbf{e}_i \mathbf{e}_j = c_{ij}^k \mathbf{e}_k, \quad c_{ij}^k \in \mathbb{C}.$$

The complex numbers c_{ij}^k , i.e., the components of the vector $\mathbf{e}_i \mathbf{e}_j$ in the basis B , are called the **structure constants** of $\mathcal{A}$.

Write down all the structure constants for the algebra of quaternions. Show that this algebra is associative.

Problem 5

Show that a quaternion is pure if and only if its square is a nonpositive real number.

Problem 6

Let $p, q \in \mathbb{H}$. Show that

- i) $(pq)^* = q^*p^*$;
- ii) $q \in \mathbb{R}$ if and only if $q^* = q$, and $q \in \mathbb{R}^3$ if and only if $q^* = -q$;
- iii) $qq^* = q^*q$ is a nonnegative real number.